

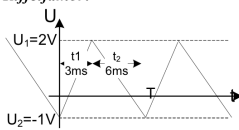
$$U = \sqrt{\frac{1}{1s} \cdot \frac{1s}{0} \left| (1,5V)^2 \cdot 0,25s + (0,5V)^2 \cdot 0,75s \right|}$$

$$U = \frac{\sqrt{3}}{2}$$

$$ff = \frac{U}{|U|} = \frac{\frac{\sqrt{3}}{2} V}{0,75V} = 1,154701$$

Repetitorium

C3. Bestimmen Sie folgende Kennwerte des folgenden Signals: Minimal- und Maximalwert, Schwingungsbreite, Scheitelwert, Gleichwert, Gleichrichtwert, Effektivwert, Schwingungsgehalt, effektive Welligkeit und Riffelfaktor!



$$U_{max} = 2V$$

$$U_{min} = -1V$$

$$U_{pp} = 2V - (-1V) = 3V$$

$$\bar{u} = \frac{1}{T} \int_0^T u(t) dt$$

$$u_1(t) \Big|_{0 \dots 3ms} = 1 \frac{V}{ms} \cdot t - 1V$$

$$u_2(t) \Big|_{3 \dots 9ms} = -0,5 \frac{V}{ms} \cdot t + 3,5V$$

$$\bar{u} = \frac{1}{9ms} \left[\int_0^{3ms} \left(1 \frac{V}{ms} t - 1V \right) dt + \int_{3ms}^{9ms} \left(-\frac{1}{2} \frac{V}{ms} t + 3,5V \right) dt \right]$$

$$\frac{1}{8 \text{ ms}} \left[\left(1 \frac{\text{V}}{\text{ms}} \cdot \frac{t^2}{2} - 1 \text{V} \cdot t \right) \Big|_0^{\frac{3}{2}} + \left(-\frac{1}{2} \frac{\text{V}}{\text{ms}} \cdot \frac{t^2}{2} + 3,5 \text{V} \cdot t \right) \Big|_3^8 \right]$$

$$\frac{1}{8 \text{ ms}} \cdot \left[\left(\frac{9 \text{ ms}^2 \text{V}}{2 \text{ ms}} - 3 \text{ msV} \right) + \left(-\frac{1}{4} \frac{81 \text{ ms}^2 \text{V}}{\text{ms}} + 3 \frac{1}{2} \text{V} \cdot 8 \text{ ms} + \frac{1}{4} \frac{\text{V}}{\text{ms}} \cdot 8 \text{ ms}^2 - 3 \frac{1}{2} \text{V} \cdot 3 \text{ ms} \right) \right]$$

$$= \frac{1}{2} \text{V}$$

$$u_{\text{eff}} = \sqrt{\frac{1}{T} \int_0^T u^2(t) dt}$$

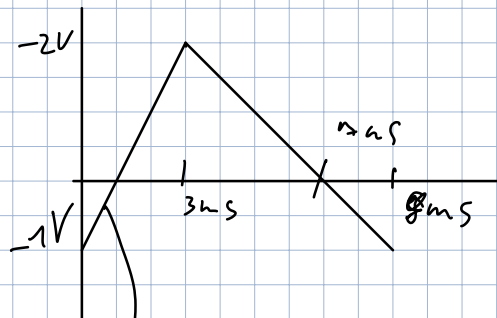
$$\int_{-1}^2 \left(1 \frac{\text{V}}{\text{ms}} \cdot t \right)^2 dt =$$

$$= 1 \frac{\text{V}^2}{\text{ms}^2} \cdot \int_{-1 \text{ ms}}^{2 \text{ ms}} t^2 dt = 1 \frac{\text{V}^2}{\text{ms}^2} \cdot \frac{t^3}{3} \Big|_{-1 \text{ ms}}^{2 \text{ ms}}$$

$$= 1 \frac{\text{V}^2}{\text{ms}^2} \cdot \left[\frac{8 \text{ ms}^3}{3} + \frac{1 \text{ ms}^3}{3} \right] = 3 \text{V}^2 \cdot \text{ms}$$

$$\int_{-4 \text{ ms}}^{2 \text{ ms}} u_2''(t) dt = + \frac{1}{4} \frac{\text{V}^2}{\text{ms}^2} \cdot \frac{t^3}{3} \Big|_{-4 \text{ ms}}^{2 \text{ ms}} = \frac{1 \text{V}^2}{4 \text{ ms}^2} \cdot \left[\frac{8}{3} + \frac{64 \text{ ms}^3}{3} \right]$$

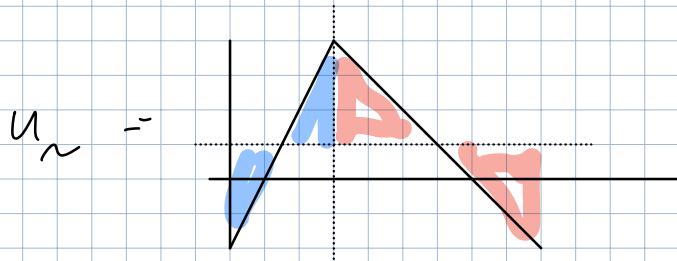
$$= 6 \text{V}^2 \cdot \text{ms}$$



$$u_1''(t) = 1 \frac{\text{V}}{\text{ms}} \cdot t \quad \begin{matrix} 2 \text{ ms} \\ -1 \text{ ms} \end{matrix}$$

$$u_2''(t) = -\frac{1}{2} \frac{\text{V}}{\text{ms}} \cdot t \quad \begin{matrix} +2 \text{ ms} \\ -4 \text{ ms} \end{matrix}$$

$$u_{eff} = \sqrt{\frac{1}{g_{ms}} \cdot (3+6) V^2 ms} = 1V$$



$$u_{\sim}(t) = u(t) - \bar{u}$$

$$u_{\sim eff} = \sqrt{\frac{1}{g_{ms}} \cdot \frac{27}{4} V^2 \cdot ms} = \frac{\sqrt{3}}{2} V$$

$$\int_{-3/2 ms}^{3/2 ms} \left| \frac{1V}{ms^2} \right|^2 \left[\frac{27g}{8 \cdot 8} + \frac{27g}{8 \cdot 8} \right] = \frac{18}{8} V^2 ms$$

$$\frac{t}{3} \int_{-3ms}^{3ms} = \frac{1}{4} \frac{V^2}{ms^2} [g + g] = \frac{18}{4} V^2 ms$$

$$u_{eff} = \sqrt{(u_{eff \sim})^2 + (\bar{u})^2} = \sqrt{\frac{3}{4} V + \frac{1}{4} V} = 1V$$

$$u_{eff \sim} = \sqrt{u_{eff}^2 - (\bar{u})^2}$$